



Capstone Lightning Talk #1

Three Project Proposals in Financial Risk

3 datasets · all classification problems

Topic 1 — Bank Customer Churn

PROBLEM

A bank earns from savings deposits. When customers go quiet and let their balance drop, the bank loses that funding — and finds out too late.

GOAL

Predict who is about to leave, so the retention team reaches them first.

EDA FINDINGS

- 28k rows, 21 columns
- Target: churn → Classification
- Not balanced: 18% churned
- 13 numerical / 5 categorical
- 3,871 missing + 3,223 hidden as text
- Outliers: current_month_credit (16%)

Topic 2 — Small Business Loans

PROBLEM

The U.S. SBA guarantees part of a small business loan. If the business fails, the bank still loses the rest — so is the risk worth taking?

GOAL

Predict if a loan is paid in full or charged off, so the officer decides with evidence.

EDA FINDINGS

- 900k rows, 27 columns
- Target: MIS_Status → Classification
- Not balanced: 17% defaulted
- 9 numerical / 8 categorical
- 747,212 missing — 734,729 in one col
- Outliers: Term (17.7%), CreateJob (16.6%)

Topic 3 — Vehicle Loan Default

PROBLEM

Lenders lose money on vehicle loan defaults. Tightening the rules also rejects good customers. The worst case is missing the very first payment.

GOAL

Predict first-payment default, so good customers are not rejected.

EDA FINDINGS

- 233k rows, 41 columns
- Target: loan_default → Classification
- Not balanced: 21% defaulted
- 24 numerical / 12 categorical
- 7,661 missings
- Outliers: PRI.CURRENT.BALANCE (17.6%)

Summary & Next Step

Metric	Topic 1 Churn	Topic 2 SB	Topic 3 Vehicle
Size	28k × 21	900k × 27	233k × 41
Problem type	Classification	Classification	Classification
Minority class	18%	17%	21%
Outliers	Yes	Yes	Yes

WHAT THEY HAVE IN COMMON

- All Classification
- All imbalanced → accuracy is the wrong metric
- All hide missing values as normal-looking values

Next step: choose one topic and build the model.

Thank you